This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

ROS-SAM: High-Quality Interactive Segmentation for Remote Sensing Moving Object

Zhe Shan[1], Yang Liu[2], Lei Zhou[1,*], Cheng Yan[3], Heng Wang[1], Xia Xie[1,*]
[1] Hainan University [2] Zhejiang University [3] Tianjin University

Abstract

The availability of large-scale remote sensing video data underscores the importance of high-quality interactive segmentation. However, challenges such as small object sizes, ambiguous features, and limited generalization make it difficult for current methods to achieve this goal. In this work, we propose ROS-SAM, a method designed to achieve high-quality interactive segmentation while preserving generalization across diverse remote sensing data. The ROS-SAM is built upon three key innovations: 1) LoRA-based fine-tuning, which enables efficient domain adaptation while maintaining SAM's generalization ability, 2) Enhancement of network deep layers to improve the discriminability of extracted features, thereby reducing misclassifications, and 3) Integration of global context with local boundary details in the mask decoder to generate high-quality segmentation masks. Additionally, we redesign the data pipeline to ensure the model learns to better handle objects at varying scales during training while focusing on high-quality predictions during inference. Experiments on remote sensing video datasets show that the data pipeline boosts the IoU by 6%, while ROS-SAM increases the IoU by 13%. Finally, when evaluated on existing remote sensing object tracking datasets, ROS-SAM demonstrates impressive zero-shot capabilities, generating masks that closely resemble manual annotations. These results confirm ROS-SAM as a powerful tool for fine-grained segmentation in remote sensing applications. Code is available at: https://github.com/ShanZard/ROS-SAM.

1. Introduction

Research into video satellite vision tasks has emerged as one of the most promising and dynamic frontiers in remote sensing today [6, 24, 25]. However, for remote sensing video moving objects (RSVMO), the small size, ambiguous features, high object density, and the complexity and cost of frame-by-frame annotation make recognizing and segmenting these objects exceedingly difficult. As illustrated in Figure 1, airplanes, cars, ships, and trains are the most common moving objects in video satellite data, and both labeling these objects and developing algorithms to detect them are formidable challenges. At the same time, the lack of pixel-level labeling information directly leads to many algorithms that cannot be trained and inferred properly.

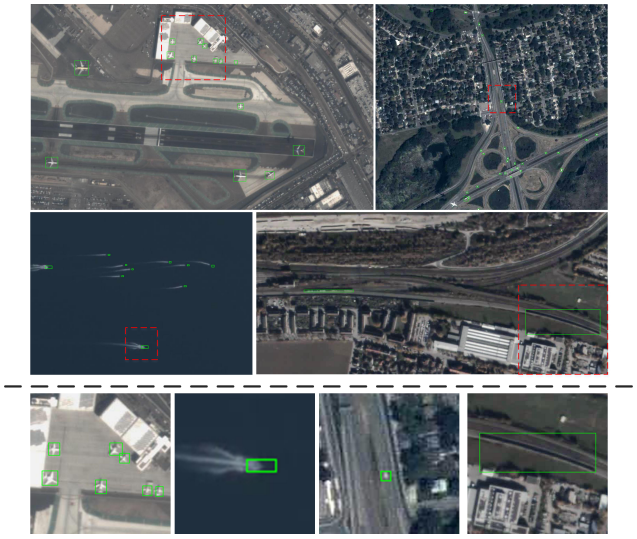


Figure 1. The recognition and annotation of these remote sensing video moving objects are particularly challenging due to factors such as small object sizes, lack of distinct features, sparse distribution, and other complexities.

Remote sensing object tracking has been actively developed for years, with many labeled datasets already available [6, 13, 26, 60, 63, 66]. It is natural to ask whether remote sensing video object segmentation can be achieved through interactive segmentation using existing object tracking datasets. Furthermore, the primary advantage of this approach lies in its ability to transform existing object tracking datasets into segmentation data, thereby advancing remote sensing video segmentation with minimal additional cost.

The Segment Anything Model (SAM) [23] is a foun-

*Corresponding author: Lei Zhou (leizhou@hainanu.edu.cn), Xia Xie (shelicy@hainanu.edu.cn)

dational model designed for visual segmentation in images, trained on 1.1 billion diverse high-quality segmentation masks. This extensive training endows SAM with powerful prior knowledge and zero-shot capabilities. Moreover, the prompt-based interactive learning strategy of SAM is well-suited for converting bounding boxes into segmentation masks. However, our experiments show that SAM is unable to achieve this conversion effectively in the context of remote sensing data. This raises the question: what distinguishes SAM's performance in general vision tasks from its limitations in remote sensing? We aim to address this question from three perspectives:

(i) Numerous empirical studies have highlighted a significant gap between general vision models and remote sensing tasks [1, 2, 33, 45, 46]. A clear example is that objects in remote sensing data are typically unaffected by gravity, and their orientation can be ambiguous. When predicting the orientation of airplanes, SAM often struggles to determine the direction the plane is facing, often predicting a generic four-pointed star. Additionally, SAM's ability to distinguish features of ground objects is limited, it struggles to differentiate between ships and waves or airplanes and boarding bridges.

(ii) SAM's prediction masks often suffer two main issues: rough mask boundaries and fragmented masks [21]. The mask decoder of SAM employs a modified Transformer architecture, which takes image embeddings and prompt tokens to generate the mask prediction. However, predicting the precise location of an object from the image embedding, which often lacks fine texture details and edge information, proves challenging. These factors make it difficult for SAM to directly produce high-quality segmentation masks that closely resemble those created through manual annotation.

(iii) SAM requires input images to be resized to a fixed resolution of 1024×1024 [20], which poses challenges when working with remote sensing data. Remote sensing images typically have large spatial dimensions, but the objects of interest are often small. As a result, downsampling can cause these objects to vanish entirely. To address these issues, a more suitable pipeline for both training and inference must be designed, one that better accommodates the unique characteristics of remote sensing imagery.

In light of these analyses, we propose an effective approach, ROS-SAM, to fine-tune SAM for **R**emote sensing video moving **O**bject **S**egmentation. In addition, we design a novel training and inference pipeline aimed at achieving high-quality segmentation. First, we use Low-Rank Adaptation (LoRA) [14] to adjust the parameters of the image encoder, minimizing the domain gap between the pre-training data and remote sensing data. We also unfreeze the last block of the image encoder to enhance its ability to extract more discriminative image embeddings. Second, we build on the HQ-SAM [21] to modify the mask decoder, integrating both early and final features from the network to improve the restoration and refinement of predictions. Furthermore, unlike HQ-SAM, we do not freeze the mask decoder of the original SAM. Instead, we apply alternating optimization to update the parameters of both the encoder and decoder. This approach strengthens SAM's applicability to RSVMO while retaining its zero-shot capabilities. Finally, we reconstruct the training and inference pipeline by incorporating large-scale jittering (LSJ) [11] and random rotations during training to capture objects at various scales. For inference, we design a center-cropping strategy based on prompts to improve segmentation quality.

The main contributions can be summarized as follows:

1. We propose ROS-SAM for achieving high-quality prediction by incorporating remote sensing special knowledge and optimizing the mask decoder.
2. We propose a novel data pipeline that introduces more multi-scale objects during training while inference focuses on high-quality inference for a single object.
3. Experimental results demonstrate that our data pipeline and ROS-SAM significantly improve performance on the SAT-MTB dataset [25], increasing IoU by 6% and 13% over the original SAM. Furthermore, ROS-SAM exhibits strong zero-shot capabilities when evaluated on existing remote sensing object tracking datasets.

2. Related Work

Video remote sensing is an emerging field within remote sensing, providing the foundation for real-time analysis of ground objects. The moving objects in remote sensing video differ significantly from those in images, exhibiting inherent characteristics such as low resolution, wide variations in scale, and high levels of noise. These factors pose significant challenges for vision algorithms in effectively analyzing the data. Early research in this field has primarily focused on three key areas: developing more robust feature extraction methods [5, 39, 40, 54], establishing temporal relationships across consecutive frames [38, 42, 57], and capturing spatiotemporal information to enable multi-frame associations [58, 64]. These efforts have significantly enhanced the accuracy of object tracking and prompted the development of this field. Video object segmentation, however, is a more practical and complex task than object tracking. It requires the model to rely on prior knowledge, significant features, or manual prompts to identify regions of interest and perform recognition across continuous frames [18, 19, 69]. Despite the challenges, researchers have made initial attempts at segmenting objects in satellite video. One notable contribution is the SAT-MTB dataset [25], a multitask benchmark that includes object tracking, detection, and segmentation tasks for satellite video. However, the dataset contains only 249 satellite videos and some of these lack annotations. With such a lim-

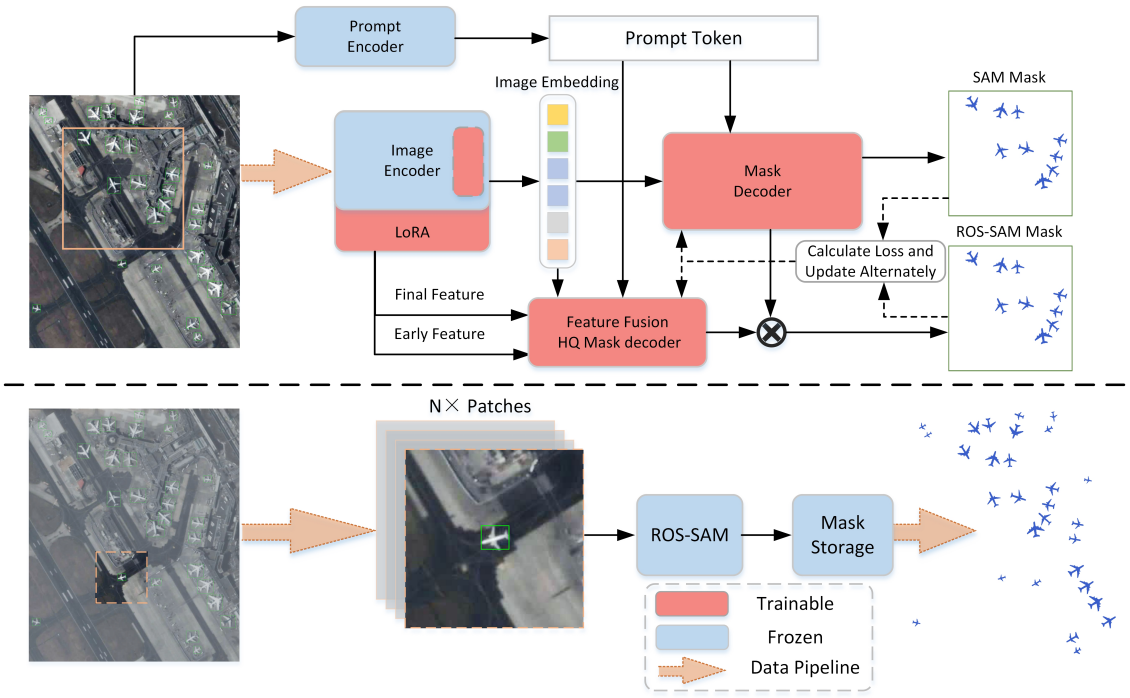


Figure 2. Overview of the proposed method. The upper part and the lower part outline the training and inference process, respectively.

ited dataset, achieving high-quality remote sensing video segmentation is virtually impossible, underscoring the need for more comprehensive and well-annotated data in this area.

Fine-tuning SAM for application across various tasks has gained significant attention from the vision community. Numerous empirical studies have demonstrated that SAM's powerful prior knowledge can effectively guide a range of downstream tasks, including medical image analysis [16, 31, 34, 70], 3D vision [7, 41, 53, 61], video analysis [10, 29, 56, 65], remote sensing [30, 32, 36, 55], and others [27, 35, 44, 62]. The key challenge in fine-tuning SAM is to preserve its robust zero-shot capabilities while adapting it to new, task-specific objectives. Two notable methods for this adaptation are Adapter and LoRA. The Adapter [3, 4] is a simple, flexible, and effective component that serves as an additional network layer, enabling the injection of task-specific guidance from a small amount of data. Its flexibility allows it to be combined with different parts of SAM, making it suitable for a variety of tasks [12, 49, 51, 67]. The LoRA [14] is a parameter efficient fine-tuning method technique that introduces a trainable linear projection layer within the Transformer architecture. This helps restore SAM's ability to extract high-level context features. Unlike Adapter, which works well for small datasets, LoRA incorporates a low-rank matrix to represent the weight change, reducing both the reliance on the pre-trained model and the risk of overfitting during fine-tuning [28, 47, 50, 59, 68]. Together, these methods provide powerful ways to adapt SAM to specialized tasks while maintaining its generalization capabilities.

3. The Proposed Method

3.1. Preliminary

We first briefly recap the design of SAM and LoRA. SAM consists of three main modules: (a) Image Encoder: a heavy ViT-based [9] backbone for image feature extraction, (b) Prompt Encoder: encoding the prompt positional information, such as points, boxes, and masks, and (c) Mask Decoder: a lightweight Transformer-based decoder that concatenates image embedding and prompt tokens for final mask prediction. The SAM model is trained on a vast large-scale dataset, which endows SAM with powerful prior knowledge and zero-shot capabilities.

LoRA is an efficient method for injecting domain-specific knowledge by freezing the pre-trained model weights and introducing small trainable decomposition matrices into each layer of the Transformer block. Specifically, given a pretrained weight matrix $W_0 \in \mathbb{R}^{m\times n}$ of SAM, LoRA adds a pair of encoder $W_e \in \mathbb{R}^{r\times n}$ and decoder $W_d \in \mathbb{R}^{m\times r}$, which represent rank decomposition matries, and $r \ll \min(m, n)$. With LoRA, the original SAM forward propagation changes from $h = W_0 x$ to

$$h = W_0 x + W_d W_e x. \tag{1}$$

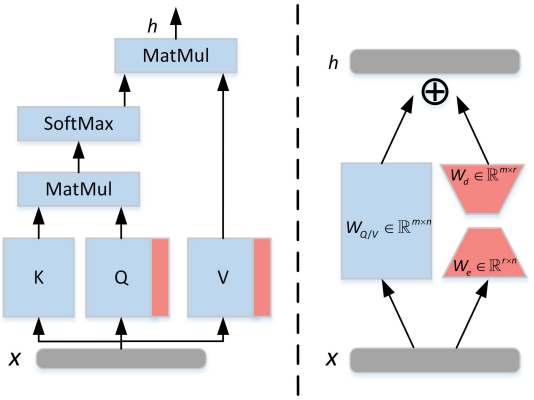


Figure 3. Illustration of introducing LoRA. Specifically, we inject low-rank decomposition matrices into the pre-trained Query and Value projection matrices of each self-attention layer.

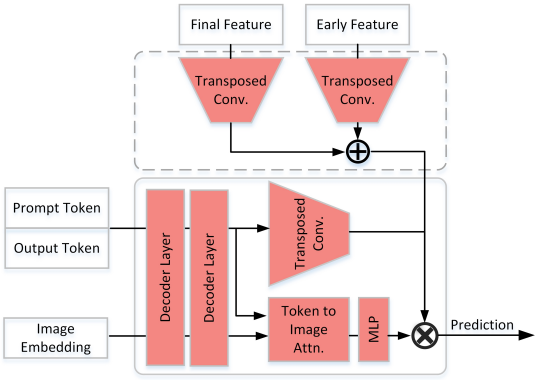


Figure 4. Illustration of the mask decoder. The solid box shows the mask decoder of the original SAM, and then splicing on the dotted box part is the HQ mask decoder.

3.2. ROS-SAM

An overview of the proposed method is depicted in Figure 2. The upper part outlines the training process of ROS-SAM. First, remote sensing images with varying sizes are fed into the image encoder using LSJ and random rotations. In the image encoder, we update the parameters of LoRA and the network's last block to incorporate remote sensing domain-specific guidance and enhance feature discrimination. Next, we exploit the HQ mask decoder, an extension of the original decoder that integrates multi-stage image features, prompt tokens, and mask tokens to produce high-quality mask predictions. Meanwhile, we introduce alternating updates to optimize the original mask decoder and the HQ mask decoder. The lower part describes the inference process. Initially, the image is cropped based on prompt information, ensuring that only one object is centered for inference. We then upsample the patches once to enlarge the features for higher-quality inference. Finally, the individual inference patches are restored to their original positions to align with the initial image, ensuring consistency with the original input.

3.2.1. Fine-tuning the image encoder

The image encoder of SAM is a heavy ViT with powerful prior knowledge. To adapt it for the RSVMO while preserving as much of this prior knowledge as possible, the key challenge lies in effectively injecting domain-specific information. To address this, we introduce LoRA to fine-tune all Transformer layers of the image encoder, as illustrated in Figure 3.

In the image encoder, the input image is first processed through patch embedding and then passed through multiple cascaded Transformer blocks within the ViT. Each Transformer block progressively captures dependencies between image patches using the self-attention mechanism. To inject domain-specific knowledge without compromising the model's original capabilities, we apply LoRA at the attention calculation stage. Specifically, we introduce a new branch parallel to the Query (Q) and Value (V) matrices. This branch decomposes the original Q and V matrices into low-rank matrices using an encoder-decoder structure, implemented via simple linear layers. During training, only the weights of the low-rank matrices are updated, allowing the model to quickly adapt to new tasks while preserving the generalizable features learned during pre-training. Based on previous experience [14], we find that updating the Q and V matrices is the most effective way to balance model performance and computational efficiency.

Additionally, we observe that the model often struggles when multiple significant objects are present within the prompt box, such as airplanes and boarding bridges. SAM is a segmentation model without explicit semantic understanding, and it tends to segment out all prominent objects, regardless of their relevance. Research [17, 71] has shown that in semantic segmentation models, shallow layers typically capture texture details, while deeper layers encode richer global context information. Building on this insight, we update the last block of the image encoder to extract more discriminative features during the training, improving the model's ability to focus on the relevant objects in complex scenes.

3.2.2. Mask decoder and high-quality mask decoder

As shown in the solid box of Figure 4, the mask decoder of the original SAM uses two Transformer layers to output the token that is adopted for mask prediction. This token predicts dynamic MLP weights and then performs point-wise products with the mask features. HQ-SAM [21] believes that an effective mask decoder should integrate both high-level object context and low-level edge information to achieve high-quality predictions. The different stages of the image encoder capture distinct types of information: the early layers typically extract local fine-grained features, while the later layers focus on global context. Based on this, we use the HQ-SAM mask decoder to generate high-quality

masks, as illustrated in Figure 4. Another key advantage of using the HQ-SAM mask decoder is that it is trained on the HQSeg-44K dataset, which contains 44,320 highly accurate mask annotations, providing strong prior knowledge of edge details.

We note two important factors: First, SAM's mask decoder is a relatively lightweight network with numerous upsampling operations involving convolutions and transposed convolutions. These operations are designed for pixel-level categorization, and updating this portion of the model does not lead to catastrophic forgetting of the prior knowledge. Second, the inductive bias inherent in convolutions and transposed convolutions requires parameter tuning for new tasks. Therefore, unlike HQ-SAM, we propose updating these weights during training. Specifically, we alternately update the SAM mask decoder and the HQ-SAM mask decoder during training to refine both components simultaneously.

3.2.3. Training and inference of ROS-SAM

ROS-SAM training. The images of RSVMO exhibit significant differences from natural images. They are typically larger in size, contain objects at varying scales, and lack a clear orientation. To address these characteristics, we redesign the training data pipeline. First, we utilize large-scale jittering [11] to enhance the model's ability to generalize across different object scales. Second, since remote sensing targets are not affected by gravity and lack fixed orientations, we introduce random rotations to generate more diverse training samples. During training, we freeze the parameters of the pre-trained SAM, making only the red components in Figure 2 learnable. We supervise the mask prediction tokens using a combination of both Binary Cross-Entropy (BCE) Loss and Dice Loss, and alternately update the network weights between the SAM Mask and ROS-SAM Mask. We trained our ROS-SAM for 48 epochs with a learning rate of 1e-3.

ROS-SAM inference. The inference pipelines used by SAM, HQ-SAM, and other works, which rely on direct resizing, are not suitable for remote sensing images, which typically have large sizes and small objects. To achieve refined predictions, we redesign the inference pipeline. First, we crop N×512×512 patches near the object based on the location information of the prompt. Next, we use bicubic interpolation to upsample the patches to N×1024×1024. These patches are then sequentially input into ROS-SAM and stored in memory. Finally, the patches are restored to the full mask based on the positional information of the prompt. Our experiments show that upsampling by a factor of two yields optimal results, while excessive upsampling leads to a significant drop in accuracy.

4. Experiments

4.1. Experimental Setup

Datasets. We use the SAT-MTB [25] for training and evaluation, which is the only available dataset for remote sensing video that includes object tracking, detection, and segmentation tasks. The SAT-MTB dataset comprises 249 videos, totaling approximately 50,000 frames, and covers the four most common types of moving objects: airplanes, cars, ships, and trains. Notably, for cars, mask generation is unnecessary, as they typically occupy only about 10 pixels and have uniform shapes that closely match the detection boxes. Additionally, some labels in the original dataset are missing, so we exclude these incomplete data points from our analysis. To mitigate the high similarity of consecutive frames, we randomly sample 1/4 of the frames from each video to create the final dataset. Detailed description of the dataset is provided in the supplemental material. Unless otherwise specified, all subsequent experiments are conducted using this modified dataset.

Evaluation metrics. To accurately quantify the performance of the model, we utilize BIoU [8] and IoU to jointly evaluate the prediction accuracy.

4.2. Ablation Experiments

We conduct comprehensive ablation studies of each proposed method, using SAM1-L as the baseline, i.e., the image encoder adopts the ViT-L version. All experimental settings remain consistent, except for the validation approach. As shown in Figure 5, each of our methods leads to a significant improvement in prediction accuracy.

In Figure 6, we visualize the prediction results, highlighting how our methods enhance the model's performance. For instance, our pipeline enables the model to focus on individual objects, resulting in extraction outcomes that are much closer to the true shape of the objects, such as the predicted airplane in the second column. Moreover, when the proposed decoder is incorporated, the model's predictions become more refined. The decoder effectively learns part of the RSVMO knowledge, improving the model's ability to discriminate between objects. This is evident in the fifth column, where SAM fails to predict the train, and the predicted shape is much larger than the train itself, likely misidentifying the railroad instead. This discrepancy can be attributed to SAM's limited domain knowledge of RSVMO. In contrast, our ROS-SAM achieves the best results, with predictions that are nearly identical to the ground truth. Notably, ROS-SAM successfully differentiates between an airplane and a boarding bridge, which all other models fail to do. Additionally, ROS-SAM delivers highly accurate predictions that closely match the true shape of objects, even when distinguishing between objects in close proximity, as shown in the third and fourth columns.

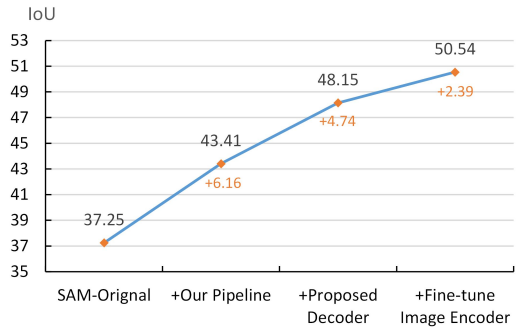


Figure 5. Ablation study of the proposed method. + indicates a new method adds to the previous one, and the orange numbers indicate the amount of growth.

Method	IoU	BIoU
SAM [23]	37.25	37.14
+inference pipeline	**43.41**	**43.30**
SAM2 [37]	36.80	36.67
+inference pipeline	**41.75**	**41.55**

Table 1. Ablation study of our inference pipeline.

While ROS-SAM excels overall, it does have some limitations. For example, it struggles to accurately differentiate the number of airplane engines and may not predict the shape of objects that are difficult to distinguish visually (where manual annotations based on prior knowledge may be more accurate). For further visual comparisons, please refer to the supplemental materials.

4.2.1. Ablation study of our pipeline

First, we validate the effectiveness of the inference pipeline using two models, SAM and SAM2, which have not been trained. As shown in Table 1, applying our inference pipeline for SAM results in an improvement of over 6% in both IoU and BIoU. For SAM2, our inference pipeline can also bring significant improvements. Next, we evaluate the impact of various factors during inference, including different sampling algorithms, sampling rates, and the effect of inferring only one object at a time. The results are shown in Table 2, the baseline configuration involves no sampling (cropped to 1024×1024 and containing multiple objects). The experiments demonstrate that all three factors have a noticeable impact on the accuracy. Among the different sampling rates, a rate of 2 proves to be the most effective, as higher upsampling rates result in jagged object edges that do not reflect the true shape of the objects. Additionally, both bicubic interpolation and limiting inference to a single object also improve the performance. Based on these findings, we adopt the best-performing configuration as the final inference pipeline.

We then conduct ablation experiments for the training pipeline of ROS-SAM. The proposed training pipeline incorporates two data augmentation techniques: LSJ (which scales objects from 0.1 to 4.0) and random rotation. As shown in Table 3, both augmentations contribute to improved model accuracy, with LSJ providing a more significant boost. This is because LSJ introduces a wider range of object scales, allowing the model to better generalize. In contrast, random rotation primarily increases the number of training samples, resulting in a more limited improvement.

Method	IoU	BIoU
Baseline	39.16	39.04
Sampling rate = 2, bilinear	41.58	41.45
Sampling rate = 2, bicubic	42.17	42.13
Sampling rate = 4, bilinear	40.37	40.11
Sampling rate = 4, bicubic	40.81	40.43
Sampling rate = 2, bicubic, only one	**43.41**	**43.30**
Sampling rate = 4, bicubic, only one	41.73	41.51

Table 2. Ablation study of the effects of different settings on inference pipeline.

Method	IoU	BIoU
ROS-SAM	49.19	49.05
+random rotation	49.43	49.26
+LSJ	50.03	49.89
+LSJ and random rotation	**50.54**	**50.36**

Table 3. Ablation study of our training pipeline.

Method	IoU	BIoU
SAM	43.41	43.30
Update mask decoder	42.82	42.69
Update HQ mask decoder	47.15	47.11
Alternately update two decoders	**48.16**	**48.03**

Table 4. Ablation study of our proposed mask decoder.

4.2.2. Ablation study of the mask decoder

The proposed mask decoder includes two contributions: 1) using the HQ mask decoder to fuse image embeddings with early texture features, and 2) alternately updating the HQ mask decoder and the original mask decoder to achieve more accurate pixel-level predictions. As shown in Table 4, directly updating the mask decoder of SAM leads to a decrease in accuracy. This is because the original decoder has already been trained on a large dataset, and directly adjusting its weights can disrupt the valuable prior knowledge it has learned. In contrast, the HQ mask decoder is a newly introduced lightweight component that integrates multiple stages of image features, resulting in a significant improvement. As discussed in Section 3.2.2, alternately updating

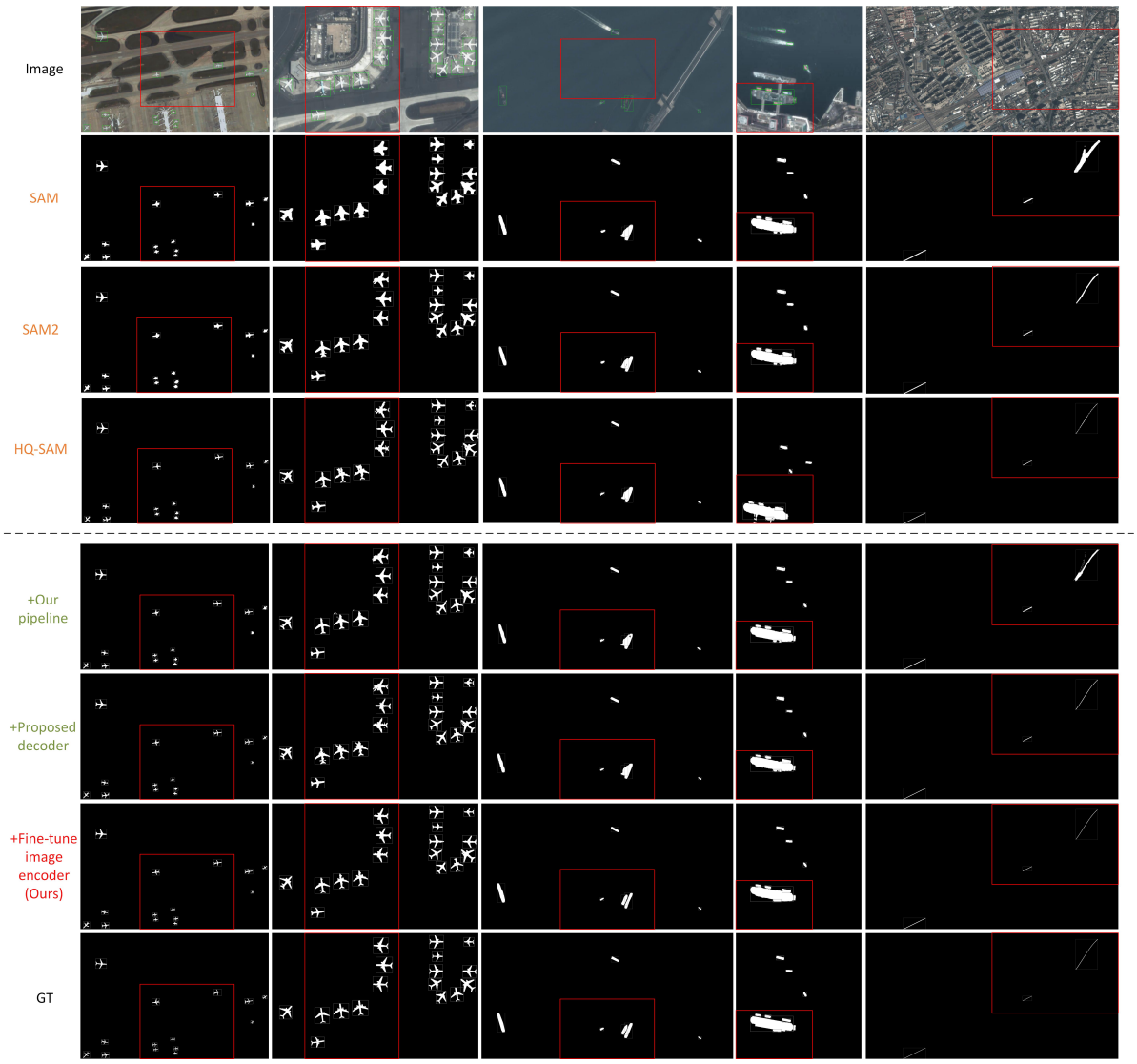


Figure 6. Visualized results of our proposed ROS-SAM and compared SOTA methods. We show the results of the ablation experiment (bottom) and the comparison experiment (top) in one image. + indicates a new method adds to the previous one.

both the HQ mask decoder and the original decoder proves to be the optimal approach, as each benefits from the other's knowledge, leading to better overall performance.

4.2.3. Ablation study of the image encoder

In this ablation experiment, we specifically evaluate the performance gains achieved by fine-tuning the image encoder using LoRA, as well as by fine-tuning only the last layer of the encoder. The experimental results, shown in Figure 7, highlight the significant performance improvements from fine-tuning the image encoder in both SAM and ROS-SAM.

4.3. Comparison with Other Methods

We conduct comparative experiments to fully validate the superiority of the ROS-SAM. As shown in Table 5, We perform a careful comparison with three algorithms, SAM, SAM2, and HQ-SAM, for which we use our inference pipeline for training and inference. We also fine-tune some other SOTA algorithms, such as [3, 15, 22, 52]. However, they all produce very poor results (the accuracy of the IoU is less than the original SAM), so we do not continue with the comparisons. Figure 6 only shows the most representative visualizations, we analyze the results of a large number of visualizations and summarize them as follows: (1) Models like SAM1 and SAM2 that have not been fine-tuned are not capable of high-quality inference, they have very fuzzy edges most of the time. (2) Although HQ-SAM generates refined prediction results, it is not semantically discriminative, so there are more misclassifications. (3) When the features are clear, the results predicted by our method are almost perfect, and when the features are fuzzy, the annotator

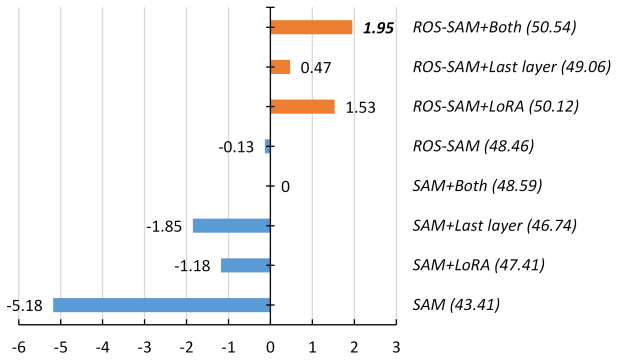


Figure 7. Ablation study of the different fine-tune methods, where SAM+Both is the baseline.

Method	IoU	BIoU
SAM [23]	43.41 (37.25)	43.30 (37.14)
SAM2 [37]	41.75 (36.80)	41.55 (36.67)
HQ-SAM [21]	47.15 (43.27)	47.11 (43.21)
Ours	**50.54**	**50.36**

Table 5. Comparison experiments with other SOTA algorithms. Results of inference according to the original algorithm configuration are shown in parentheses.

can predict the approximate shape of the target, which is difficult to achieve with our method.

4.4. Experiment Results on Other Datasets

4.4.1. Segmentation results on static image dataset

Method	iSAID		NPWS VHR-10	
	IoU	BIoU	IoU	BIoU
SAM	53.19	48.86	65.54	68.44
SAM2	55.79	52.34	68.84	69.18
HQ-SAM	63.96	57.48	78.44	76.10
Ours	**73.22**	**68.22**	**87.46**	**84.56**

Table 6. The results on static remote sensing datasets.

We have conducted experiments on two static datasets to further verify the effectiveness and generalization of the proposed method. iSAID [48] and NPWS VHR-10 [43] datasets are representative instance segmentation datasets, comprising 15 and 10 common types of objects, respectively. As shown in Table 6, our method is also applicable to static images and has more advantages.

4.4.2. Zero-shot segmentation on object tracking

The zero-shot capabilities of the model are crucial for our task, and we evaluate its performance on several remote sensing object tracking datasets, including SatSOT [66], VISO [60], and OOTB [60]. These datasets differ from the training set in terms of resolution, sensor type, and the time and location of capture, providing a robust test of the model's generalization ability. Since the tracking datasets lack segmentation masks, we conduct a qualitative evaluation rather than a quantitative analysis of the predictions. As shown in Figure 8, ROS-SAM demonstrates superior performance across these datasets, consistently generating high-quality segmentation masks. Due to the length limitation of the paper, more visualization results refer to the supplemental materials.

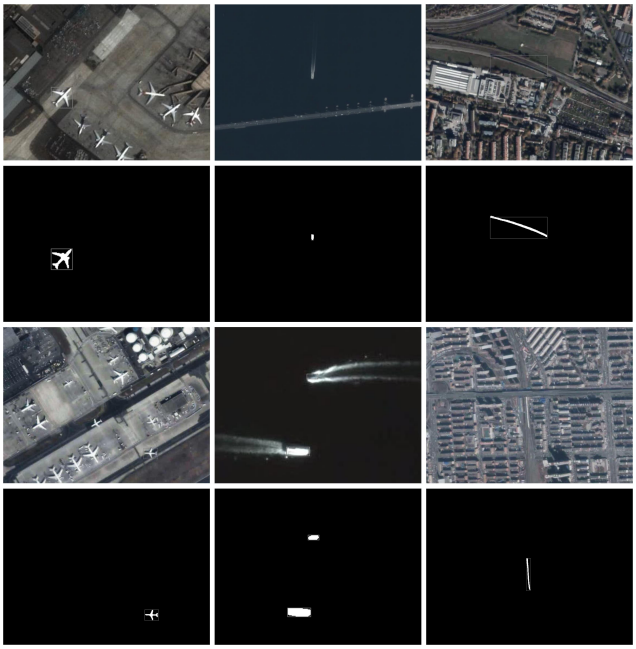


Figure 8. Results on remote sensing object tracking dataset.

5. Conclusion

In this paper, we present ROS-SAM, a model built upon SAM with contributions to three critical components: the data pipeline, image encoder, and mask decoder. These components are specifically designed to achieve high-quality mask prediction in remote sensing moving object segmentation. Ablation studies and comparative experiments show that these modifications substantially enhance SAM's performance in RSVMO tasks. Additionally, the generalization ability of ROS-SAM is validated through cross-dataset experiments. We anticipate that ROS-SAM will become a powerful tool for advancing fine-grained segmentation in remote sensing video analysis.

Acknowledgement. This work was supported by the National Natural Science Foundation of China (No. 62402354, No. 62362023), and the Key Research and Development Project of Hainan Province (ZDYF2024GXJS313, ZDYF2024GXJS262), and Hainan Province Science and Technology Special Fund (ATIC-202302001).

References

[1] Favyen Bastani, Piper Wolters, Ritwik Gupta, Joe Ferdinando, and Aniruddha Kembhavi. Satlaspretrain: A large-scale dataset for remote sensing image understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16772–16782, 2023. 2

[2] Keumgang Cha, Junghoon Seo, and Taekyung Lee. A billion-scale foundation model for remote sensing images. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, pages 1–17, 2024. 2

[3] Tianrun Chen, Lanyun Zhu, Chaotao Deng, Runlong Cao, Yan Wang, Shangzhan Zhang, Zejian Li, Lingyun Sun, Ying Zang, and Papa Mao. Sam-adapter: Adapting segment anything in underperformed scenes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pages 3367–3375, 2023. 3, 7

[4] Tianrun Chen, Ankang Lu, Lanyun Zhu, Chaotao Ding, Chunan Yu, Deyi Ji, Zejian Li, Lingyun Sun, Papa Mao, and Ying Zang. Sam2-adapter: Evaluating & adapting segment anything 2 in downstream tasks: Camouflage, shadow, medical image segmentation, and more. *arXiv preprint arXiv:2408.04579*, 2024. 3

[5] Yuzeng Chen, Yuqi Tang, Zhiyong Yin, Te Han, Bin Zou, and Huihui Feng. Single object tracking in satellite videos: A correlation filter-based dual-flow tracker. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15:6687–6698, 2022. 2

[6] Yuzeng Chen, Yuqi Tang, Yi Xiao, Qiangqiang Yuan, Yuwei Zhang, Fengqing Liu, Jiang He, and Liangpei Zhang. Satellite video single object tracking: A systematic review and an oriented object tracking benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing*, 210:212–240, 2024. 1

[7] Zhimin Chen, Liang Yang, Yingwei Li, Longlong Jing, and Bing Li. Sam-guided masked token prediction for 3d scene understanding. *arXiv preprint arXiv:2410.12158*, 2024. 3

[8] Bowen Cheng, Ross Girshick, Piotr Dollar, Alexander C. Berg, and Alexander Kirillov. Boundary iou: Improving object-centric image segmentation evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15334–15342, 2021. 5

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. 3

[10] Hao Fang, Tong Zhang, Xiaofei Zhou, and Xinxin Zhang. Learning better video query with sam for video instance segmentation. *IEEE Transactions on Circuits and Systems for Video Technology*, pages 1–1, 2024. 3

[11] Golnaz Ghiasi, Yin Cui, Aravind Srinivas, Rui Qian, Tsung-Yi Lin, Ekin D Cubuk, Quoc V Le, and Barret Zoph. Simple copy-paste is a strong data augmentation method for instance segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2918–2928, 2021. 2, 5

[12] Shizhan Gong, Yuan Zhong, Wenao Ma, Jinpeng Li, Zhao Wang, Jingyang Zhang, Pheng-Ann Heng, and Qi Dou. 3dsam-adapter: Holistic adaptation of sam from 2d to 3d for promptable tumor segmentation. *Medical Image Analysis*, 98:103324, 2024. 3

[13] Qibin He, Xian Sun, Zhiyuan Yan, Beibei Li, and Kun Fu. Multi-object tracking in satellite videos with graph-based multitask modeling. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–13, 2022. 1

[14] Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. 2, 3, 4

[15] Xinrong Hu, Xiaowei Xu, and Yiyu Shi. How to efficiently adapt large segmentation model (sam) to medical images. *arXiv preprint arXiv:2306.13731*, 2023. 7

[16] Yuhao Huang, Xin Yang, Lian Liu, Han Zhou, Ao Chang, Xinrui Zhou, Rusi Chen, Junxuan Yu, Jiongquan Chen, Chaoyu Chen, Sijing Liu, Haozhe Chi, Xindi Hu, Kejuan Yue, Lei Li, Vicente Grau, Deng-Ping Fan, Fajin Dong, and Dong Ni. Segment anything model for medical images? *Medical Image Analysis*, 92:103061, 2024. 3

[17] Deyi Ji, Haoran Wang, Mingyuan Tao, Jianqiang Huang, Xian-Sheng Hua, and Hongtao Lu. Structural and statistical texture knowledge distillation for semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16876–16885, 2022. 4

[18] Licheng Jiao, Ruohan Zhang, Fang Liu, Shuyuan Yang, Biao Hou, Lingling Li, and Xu Tang. New generation deep learning for video object detection: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 33(8): 3195–3215, 2022. 2

[19] Licheng Jiao, Xin Zhang, Xu Liu, Fang Liu, Shuyuan Yang, Wenping Ma, Lingling Li, Puhua Chen, Zhixi Feng, Yuwei Guo, Xu Tang, Biao Hou, Xiangrong Zhang, Jing Bai, Dou Quan, and Junpeng Zhang. Transformer meets remote sensing video detection and tracking: A comprehensive survey. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16:1–45, 2023. 2

[20] Sota Kato, Hinako Mitsuoka, and Kazuhiro Hotta. Generalized sam: Efficient fine-tuning of sam for variable input image sizes. *arXiv preprint arXiv:2408.12406*, 2024. 2

[21] Lei Ke, Mingqiao Ye, Martin Danelljan, Yifan liu, Yu-Wing Tai, Chi-Keung Tang, and Fisher Yu. Segment anything in high quality. In *Advances in Neural Information Processing Systems*, pages 29914–29934, 2023. 2, 4, 8

[22] Joohyeok Kim, Joonhyeon Song, Seohwan Yun, Seongho Yoon, and Sangmin Lee. Cad: Memory efficient convolutional adapter for segment anything. *arXiv preprint arXiv:2409.15889*, 2024. 7

[23] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollar, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026, 2023. 1, 6, 8

[24] Shengyang Li, Xian Sun, Yanfeng Gu, Yixuan Lv, Manqi Zhao, Zhuang Zhou, Weilong Guo, Yuhan Sun, Han Wang, and Jian Yang. Recent advances in intelligent processing of satellite video: Challenges, methods, and applications. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16:6776–6798, 2023. 1

[25] Shengyang Li, Zhuang Zhou, Manqi Zhao, Jian Yang, Weilong Guo, Yixuan Lv, Longxuan Kou, Han Wang, and Yanfeng Gu. A multitask benchmark dataset for satellite video: Object detection, tracking, and segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–21, 2023. 1, 2, 5

[26] Yuxuan Li, Licheng Jiao, Zhongjian Huang, Xin Zhang, Ruohan Zhang, Xue Song, Chenxi Tian, Zixiao Zhang, Fang Liu, Shuyuan Yang, Biao Hou, Wenping Ma, Xu Liu, and Lingling Li. Deep learning-based object tracking in satellite videos: A comprehensive survey with a new dataset. *IEEE Geoscience and Remote Sensing Magazine*, 10(4):181–212, 2022. 1

[27] Yang Liu, Muzhi Zhu, Hengtao Li, Hao Chen, Xinlong Wang, and Chunhua Shen. Matcher: Segment anything with one shot using all-purpose feature matching. In *The Twelfth International Conference on Learning Representations*, 2024. 3

[28] Xiaoyan Lu and Qihao Weng. Multi-lora fine-tuned segment anything model for urban man-made object extraction. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–19, 2024. 3

[29] Zhihe Lu, Zeyu Xiao, Jiawang Bai, Zhiwei Xiong, and Xinchao Wang. Can sam boost video super-resolution? *arXiv preprint arXiv:2305.06524*, 2023. 3

[30] Muying Luo, Tao Zhang, Shiqing Wei, and Shunping Ji. Sam-rsis: Progressively adapting sam with box prompting to remote sensing image instance segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–14, 2024. 3

[31] Jun Ma, Yuting He, Feifei Li, Lin Han, Chenyu You, and Bo Wang. Segment anything in medical images. *Nature Communications*, 15(1):654, 2024. 3

[32] Xianping Ma, Qianqian Wu, Xingyu Zhao, Xiaokang Zhang, Man-On Pun, and Bo Huang. Sam-assisted remote sensing imagery semantic segmentation with object and boundary constraints. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–16, 2024. 3

[33] Oscar Mañas, Alexandre Lacoste, Xavier Giró-i Nieto, David Vazquez, and Pau Rodríguez. Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9414–9423, 2021. 2

[34] Maciej A. Mazurowski, Haoyu Dong, Hanxue Gu, Jichen Yang, Nicholas Konz, and Yixin Zhang. Segment anything model for medical image analysis: An experimental study. *Medical Image Analysis*, 89:102918, 2023. 3

[35] Shentong Mo and Yapeng Tian. Av-sam: Segment anything model meets audio-visual localization and segmentation. *arXiv preprint arXiv:2305.01836*, 2023. 3

[36] Lucas Prado Osco, Qiusheng Wu, Eduardo Lopes de Lemos, Wesley Nunes Gonçalves, Ana Paula Marques Ramos, Jonathan Li, and José Marcato. The segment anything model (sam) for remote sensing applications: From zero to one shot. *International Journal of Applied Earth Observation and Geoinformation*, 124:103540, 2023. 3

[37] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 6, 8

[38] Jia Shao, Bo Du, Chen Wu, and Yan Pingkun. Pasiam: Predicting attention inspired siamese network, for space-borne satellite video tracking. In *2019 IEEE International Conference on Multimedia and Expo*, pages 1504–1509. IEEE, 2019. 2

[39] Jia Shao, Bo Du, Chen Wu, and Lefei Zhang. Can we track targets from space? a hybrid kernel correlation filter tracker for satellite video. *IEEE Transactions on Geoscience and Remote Sensing*, 57(11):8719–8731, 2019. 2

[40] Jia Shao, Bo Du, Chen Wu, and Lefei Zhang. Tracking objects from satellite videos: A velocity feature based correlation filter. *IEEE Transactions on Geoscience and Remote Sensing*, 57(10):7860–7871, 2019. 2

[41] Yiqing Shen, Jingxing Li, Xinyuan Shao, Blanca Inigo Romillo, Ankush Jindal, David Dreizin, and Mathias Unberath. Fastsam3d: An efficient segment anything model for 3d volumetric medical images. In *Medical Image Computing and Computer Assisted Intervention*, pages 542–552, 2024. 3

[42] Wei Song, Licheng Jiao, Fang Liu, Xu Liu, Lingling Li, Shuyuan Yang, Biao Hou, and Wenhua Zhang. A joint siamese attention-aware network for vehicle object tracking in satellite videos. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–17, 2022. 2

[43] Hao Su, Shunjun Wei, Shan Liu, Jiadian Liang, Chen Wang, Jun Shi, and Xiaoling Zhang. Hq-isnet: High-quality instance segmentation for remote sensing imagery. *Remote Sensing*, 12(6):989, 2020. 8

[44] Xiaorui Sun, Jun Liu, Heng Tao Shen, Xiaofeng Zhu, and Ping Hu. On efficient variants of segment anything model: A survey. *arXiv preprint arXiv:2410.04960*, 2024. 3

[45] Chao Tao, Ji Qi, Mingning Guo, Qing Zhu, and Haifeng Li. Self-supervised remote sensing feature learning: Learning paradigms, challenges, and future works. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–26, 2023. 2

[46] Di Wang, Jing Zhang, Bo Du, Gui-Song Xia, and Dacheng Tao. An empirical study of remote sensing pretraining. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–20, 2023. 2

[47] Xuehao Wang, Feiyang Ye, and Yu Zhang. Task-aware low-rank adaptation of segment anything model. *arXiv preprint arXiv:2403.10971*, 2024. 3

[48] Syed Waqas Zamir, Aditya Arora, Akshita Gupta, Salman Khan, Guolei Sun, Fahad Shahbaz Khan, Fan Zhu, Ling Shao, Gui-Song Xia, and Xiang Bai. isaid: A large-scale dataset for instance segmentation in aerial images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pages 28–37, 2019. 8

[49] Junde Wu, Wei Ji, Yuanpei Liu, Huazhu Fu, Min Xu, Yanwu Xu, and Yueming Jin. Medical sam adapter: Adapting segment anything model for medical image segmentation. *arXiv preprint arXiv:2304.12620*, 2023. 3

[50] Aoran Xiao, Weihao Xuan, Heli Qi, Yun Xing, Naoto Yokoya, and Shijian Lu. Segment anything with multiple modalities. *arXiv preprint arXiv:2408.09085*, 2024. 3

[51] Zhaozhi Xie, Bochen Guan, Weihao Jiang, Muyang Yi, Yue Ding, Hongtao Lu, and Lei Zhang. Pa-sam: Prompt adapter sam for high-quality image segmentation. *2024 IEEE International Conference on Multimedia and Expo*, 2024. 3

[52] Xinyu Xiong, Zihuang Wu, Shuangyi Tan, Wenxue Li, Feilong Tang, Ying Chen, Siying Li, Jie Ma, and Guanbin Li. Sam2-unet: Segment anything 2 makes strong encoder for natural and medical image segmentation. *arXiv preprint arXiv:2408.08870*, 2024. 7

[53] Xiuwei Xu, Huangxing Chen, Linqing Zhao, Ziwei Wang, Jie Zhou, and Jiwen Lu. Embodiedsam: Online segment any 3d thing in real time. *arXiv preprint arXiv:2408.11811*, 2024. 3

[54] Shiyu Xuan, Shengyang Li, Zifei Zhao, Zhuang Zhou, Wanfeng Zhang, Hong Tan, Guisong Xia, and Yanfeng Gu. Rotation adaptive correlation filter for moving object tracking in satellite videos. *Neurocomputing*, 438:94–106, 2021. 2

[55] Zhiyuan Yan, Junxi Li, Xuexue Li, Ruixue Zhou, Wenkai Zhang, Yingchao Feng, Wenhui Diao, Kun Fu, and Xian Sun. Ringmo-sam: A foundation model for segment anything in multimodal remote-sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–16, 2023. 3

[56] Jinyu Yang, Mingqi Gao, Zhe Li, Shang Gao, Fangjing Wang, and Feng Zheng. Track anything: Segment anything meets videos. *arXiv preprint arXiv:2304.11968*, 2023. 3

[57] Jianwei Yang, Zongxu Pan, Ziming Wang, Bin Lei, and Yuxin Hu. Siammdm: an adaptive fusion network with dynamic template for real-time satellite video single object tracking. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–19, 2023. 2

[58] Xiaoyan Yang, Licheng Jiao, Yangyang Li, Xu Liu, Lingling Li, Puhua Chen, Fang Liu, and Shuyuan Yang. High-order relation learning transformer for satellite video object tracking. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–15, 2024. 2

[59] Zehao Ye, Lucy Lovell, Asaad Faramarzi, and Jelena Ninić. Sam-based instance segmentation models for the automation of structural damage detection. *Advanced Engineering Informatics*, 62:102826, 2024. 3

[60] Qian Yin, Qingyong Hu, Hao Liu, Feng Zhang, Yingqian Wang, Zaiping Lin, Wei An, and Yulan Guo. Detecting and tracking small and dense moving objects in satellite videos: A benchmark. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–18, 2022. 1, 8

[61] Yingda Yin, Yuzheng Liu, Yang Xiao, Daniel Cohen-Or, Jingwei Huang, and Baoquan Chen. Sai3d: Segment any instance in 3d scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3292–3302, 2024. 3

[62] Chenshuang Zhang, Chaoning Zhang, Taegoo Kang, Donghun Kim, Sung-Ho Bae, and In So Kweon. Attack-sam: Towards attacking segment anything model with adversarial examples. *arXiv preprint arXiv:2305.00866*, 2023. 3

[63] Wenhua Zhang, Licheng Jiao, Fang Liu, Lingling Li, Xu Liu, and Jia Liu. Mblt: Learning motion and background for vehicle tracking in satellite videos. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–15, 2022. 1

[64] Xin Zhang, Licheng Jiao, Lingling Li, Xu Liu, Fang Liu, Wenping Ma, and Shuyuan Yang. Adaptive multi-scale transformer tracker for satellite videos. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–16, 2024. 2

[65] Zhenghao Zhang, Zhichao Wei, Shengfan Zhang, Zuozhuo Dai, and Siyu Zhu. Uvosam: A mask-free paradigm for unsupervised video object segmentation via segment anything model. *arXiv preprint arXiv:2305.12659*, 2023. 3

[66] Manqi Zhao, Shengyang Li, Shiyu Xuan, Longxuan Kou, Shuai Gong, and Zhuang Zhou. Satsot: A benchmark dataset for satellite video single object tracking. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–11, 2022. 1, 8

[67] Linghao Zheng, Xinyang Pu, and Feng Xu. Tuning a sam-based model with multi-cognitive visual adapter to remote sensing instance segmentation. *arXiv preprint arXiv:2408.08576*, 2024. 3

[68] Zihan Zhong, Zhiqiang Tang, Tong He, Haoyang Fang, and Chun Yuan. Convolution meets lora: Parameter efficient finetuning for segment anything model. In *The Twelfth International Conference on Learning Representations*, 2024. 3

[69] Tianfei Zhou, Fatih Porikli, David J. Crandall, Luc Van Gool, and Wenguan Wang. A survey on deep learning technique for video segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(6):7099–7122, 2023. 2

[70] Jiayuan Zhu, Yunli Qi, and Junde Wu. Medical sam 2: Segment medical images as video via segment anything model 2. *arXiv preprint arXiv:2408.00874*, 2024. 3

[71] Lanyun Zhu, Deyi Ji, Shiping Zhu, Weihao Gan, Wei Wu, and Junjie Yan. Learning statistical texture for semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12537–12546, 2021. 4